MERIDIAN PETROCHEMICAL PLANT

Pump P-101 Maintenance Manual
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1. Equipment Overview

Pump P-101 is a single-stage centrifugal pump manufactured by AquaFlow Industries, model AF-3500C. It is installed in Production Unit 3 of the Meridian Petrochemical Plant and is critical for feedstock transfer operations. The pump was commissioned in March 2019 and operates continuously during production cycles.

The pump operates at a nominal speed of 3000 RPM driven by a 75 kW electric motor through a flexible coupling. Design flow rate is 150 cubic meters per hour with a total dynamic head of 45 meters. The pump handles light hydrocarbon feedstock at temperatures between 25 and 65 degrees Celsius.

Major components include: the impeller (316L stainless steel, closed type), the mechanical seal (John Crane Type 2100, single cartridge), the bearing housing with two SKF 6205-2RS deep groove ball bearings, the shaft (AISI 4140 steel), and the volute casing (cast iron Grade 25). The pump is mounted on a concrete foundation with vibration isolation pads.

2. Operating Parameters

Normal operating parameters must be maintained within the following limits at all times. Inlet pressure: 2.0 to 4.0 bar gauge. Outlet pressure: 8.0 to 12.0 bar gauge. Flow rate: 120 to 180 cubic meters per hour. Bearing temperature: not to exceed 80 degrees Celsius. Vibration level: not to exceed 4.5 mm/s RMS on bearing housing.

The pump must not be operated below minimum continuous stable flow of 60 cubic meters per hour to prevent recirculation damage. Dry running for more than 30 seconds will cause catastrophic mechanical seal failure and is strictly prohibited.

3. Preventive Maintenance Schedule

Daily Checks: Verify bearing temperature using infrared thermometer. Check vibration levels with portable analyzer. Inspect mechanical seal for visible leakage. Verify inlet and outlet pressure readings on local gauges. Record all readings in the daily logbook.

Weekly Maintenance: Grease motor bearings using Shell Gadus S2 V220 grease (2 pumps per grease point). Check coupling alignment using dial indicators. Inspect foundation bolts for tightness. Clean strainer basket on the suction line.

Monthly Maintenance: Perform vibration spectrum analysis and compare to baseline. Check motor current draw against nameplate rating. Inspect piping supports and expansion joints. Verify all safety instrumentation (pressure relief valve PSV-101, flow switch FS-101).

Semi-Annual Overhaul: Replace mechanical seal assembly. Replace both SKF 6205-2RS bearings. Inspect impeller for erosion and cavitation damage. Measure shaft runout (max 0.05 mm TIR). Replace coupling elastomer element. Perform hydrostatic test at 1.5x MAWP.

Annual Inspection: Full disassembly and inspection of all wetted parts. Measure casing wear ring clearance (replace if exceeds 0.5 mm). NDT inspection of shaft for cracks. Thickness measurement of volute casing. Review and update maintenance procedures as needed.

4. Troubleshooting Guide

Symptom: Excessive Vibration (above 4.5 mm/s RMS). Possible Causes: Bearing failure or degradation, shaft misalignment, impeller imbalance due to erosion, loose foundation bolts, cavitation. Corrective Actions: Immediately reduce pump speed to 2000 RPM. Perform vibration spectrum analysis to identify the dominant frequency. If bearing defect frequency is present, schedule bearing replacement within 48 hours. Check and correct alignment using reverse dial indicator method. Verify inlet pressure is above minimum NPSH required.

Symptom: High Bearing Temperature (above 80 degrees Celsius). Possible Causes: Insufficient or degraded lubrication, bearing overload due to misalignment, internal bearing damage, ambient temperature effects. Corrective Actions: Check grease condition and re-lubricate if dry or contaminated. Verify alignment is within tolerance (0.05 mm angular, 0.08 mm offset). If temperature exceeds 95 degrees Celsius, shut down immediately and replace bearings.

Symptom: Reduced Flow Rate (below 120 cubic meters per hour). Possible Causes: Impeller wear or erosion damage, suction strainer blockage, air entrainment in suction line, worn wear rings increasing internal recirculation. Corrective Actions: Check and clean suction strainer. Verify suction line is free of air leaks. Measure actual head versus curve to determine impeller condition. If performance has degraded more than 10 percent from baseline, schedule impeller replacement.

Symptom: Mechanical Seal Leakage. Possible Causes: Seal face damage, shaft sleeve scoring, incorrect seal setting, thermal shock from rapid temperature changes, excessive shaft deflection. Corrective Actions: Minor drip leakage (less than 5 ml per hour) can be monitored. Continuous stream leakage requires immediate shutdown. Replace the complete John Crane Type 2100 cartridge seal assembly. Inspect shaft sleeve for scoring and replace if runout exceeds 0.025 mm.

Symptom: Abnormal Noise During Operation. Possible Causes: Cavitation (crackling or rattling sound), bearing damage (grinding or squealing), impeller rubbing on wear ring, loose internal components. Corrective Actions: For cavitation noise, increase suction pressure or reduce flow rate. For bearing noise, schedule immediate replacement. For rubbing noise, shut down and check wear ring clearance. Document all noise events with timestamp and operating conditions.